<u>OSSP (Operating Systems And Systems Programming) - 25CS2104E</u>

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Section: 8

PRACTICAL-2:

Aim: Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.

```
venkatsaikrishna5@VSK-HP:~$ cd ~
venkatsaikrishna5@VSK-HP:~$ mkdir practical02
mkdir: practical02: File exists
venkatsaikrishna5@VSK-HP:~$ cd practical02
venkatsaikrishna5@VSK-HP:~/practical02$ pwd
/home/venkatsaikrishna5/practical02
venkatsaikrishna5@VSK-HP:~/practical02$ echo "Hello World" > sa
mple.txt
venkatsaikrishna5@VSK-HP:~/practical02$ echo "This is my operat
ing systems assignment." >> sample.txt
echo "This file will be copied using system calls." >> sample.t
xt
venkatsaikrishna5@VSK-HP:~/practical02$ cat sample.txt
Hello World
This is my operating systems assignment.
This file will be copied using system calls.
venkatsaikrishna5@VSK-HP:~/practical02$ nano copy.c
venkatsaikrishna5@VSK-HP:~/practical02$ gcc copy.c -o copy
venkatsaikrishna5@VSK-HP:~/practical02$ ./copy
File copied successfully.
venkatsaikrishna5@VSK-HP:~/practical02$ cat copy.txt
Hello World
This is my operating systems assignment.
This file will be copied using system calls.
venkatsaikrishna5@VSK-HP:~/practical02$ diff sample.txt copy.tx
t
```

$ strace cat sample.txt

```
cxcc,1c(                [
!I1·1,(  )                         =
"' :,,p(
         ) =
:,c' ccss (                  )      =
opernt(
t c, 1·"- t( {st_mode=       st size=        }) =
1·1··:,w                                 ) =
clcsc(                             =
opernt
               ) =
1·c,HI(
fst:,t( {st_mode=       st size=        }) =
lii  ap(                                     )
=
III ··,I1J(
                      ) =
I'I  ·,,p(
                 =
1·1· a;(
                      ) =
I"I··c.p(
                   =
clcsc•(                          =
opcrnt(
               =
:·c acl(
               ) =
h1.-;t( {st_mode=       st size=        }) =
!·1· d[J(                                    )
```

```
mmap(0x753e82e0b000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|M
AP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x753e82e0b000
mmap(0x753e82e2f000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MA
P_DENYWRITE, 3, 0x28000) = 0x753e82e2f000
mmap(0x753e82e33000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MA
P_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x753e82e33000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONL
Y|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0
\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0)
 = 0x753e82ce1000
mmap(0x753e82cf3000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|M
AP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x753e82cf3000
mmap(0x753e82d93000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|M
AP_DENYWRITE, 3, 0xb2000) = 0x753e82d93000
mmap(0x753e82e05000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MA
P_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x753e82e05000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONL
Y|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\
0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0
\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0
\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0)
 = 0x753e82a00000
mmap(0x753e82a28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|
MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x753e82a28000
mmap(0x753e82bc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|M
AP_DENYWRITE, 3, 0x1c0000) = 0x753e82bc0000
```

```
strace -o trace.txt cat
sample.txt
Hello World
This is my operating systems assignment.
This file will be copied using system calls.
$ grep -E "openlreadlwrit
elcloselexecvelexit" trace.txt
("/usr/bin/cat", ["cat", "sample.txt"], 0x7fff9f93df08 /*
26 vars*/) = 0
at(AT_FDCWD, "/etc/ld.so.cache", O_RDONLYIO_CLOEXEC) = 3
(3) = 0
at(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O
_RDONLYIO_CLOEXEC) = 3
(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0
\0\0\0\0"..., 832) = 832
(3) = 0
at(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_R
DONLYIO_CLOEXEC) = 3
(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0
\0\0\0\0"..., 832) = 832
(3) = 0
at(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONL
YIO_CLOEXEC) = 3
(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0
\0\0\0\0"..., 832) = 832
(3) = 0
at(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONL
YIO_CLOEXEC) = 3
(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\
0\0\0\0\0"..., 832) = 832
p 64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0
\0\0\0\0\0"..., 840, 64) = 840
p 64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0
\0\0\0\0\0"..., 840, 64) = 840
(3) = 0

3 = 0
at(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O
_RDONLYjO_CLOEXEC) = 3
(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0
\0\0\0\0"..., 832) = 832
(3) = 0
at(AT_FDCWD, "/proc/filesystems", O_RDONLYIO_CLOEXEC) = 3
(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 10211) = 442
(3) = 0
at(AT_FDCWD, "/proc/mounts", O_RDONLYjO_CLOEXEC) = 3
(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
(3, "e 0 0\ndevpts /dev/pts devpts rw,"..., 10211) = 10211
(3, "c,relatime 0 0\nhugetlbfs /dev/hu"..., 10211) = 10211
(3, "s /mnt/wslg/run/user/1000 tmpfs "..., 10211) = 118
(3, "", 10211) =0
(3) = 0
at(AT_FDCWD, "/proc/self/maps", O_RDONLYjO_CLOEXEC) = 3
(3, "5ad9ebbff000-5ad9ebcf1000 r--p 0"..., 10211) = 10211
(3, "3bfe000-7838d3bff000 r--p 000a90"..., 10211) = 10211
(3, "00 08:30 13650 ..., 1024) = 1024
(3, " /usr/lib/x86_611-lin"..., 10211) = 10211
(3, "0 136llll /us"..., 10211) = 566
(3) = 0
at(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl
", O_RDONLYIO_CLOEXEC) = 3
(3, "ll Common strings shared across a"..., 11080) = 11080
(3, "", 32) =0
(3) = 0
at(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl",
O_RDONLYjO_CLOEXEC) = -1 ENOENT (No such file or directory)
at(AT_FDCWD, "sample.txt", O_RDONLYjO_CLOEXEC) = 3
(5) = 0
(11) = 0
(3) = 0
_group(0) =?
+++ ed with 0 +++
```

```
cat sample.txt
Hello World
This is my operating systems assignment.
This file will be co ied usin  s stem calls.
```

System Call	What it does
execve()	Loads and starts the cat program
openat()	Opens sample.txt
read()	Reads data from the file
write()	Displays data on the terminal
close()	Closes the file descriptor
exit_group()	Terminates the process

Kernel services involved

1. **Process Management**

execve() and exit_group() are related to process execution and termination.

2. **File Management**

openat() and close() manage file descriptors and files.

3. **1/0 Management**

read() and write() handle data transfer between the program and the kernel.

Code:

```
  GNU nano 8.7.1                    copy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;

    // Open source file
    source = open("sample.txt", O_RDONLY);

    if (source == -1)
    {
        perror("Error opening sample.txt");
        return 1;
    }

    // Open/Create destination file
    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUN>

    if (destination == -1)
    {
        perror("Error opening copy.txt");
        close(source);
        return 1;
    }

    // Copy contents
    while ((bytesRead = read(source, buffer, sizeof(buffer))) >
    {
        write(destination, buffer, bytesRead);
    }
```

```
    // Close files
    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```